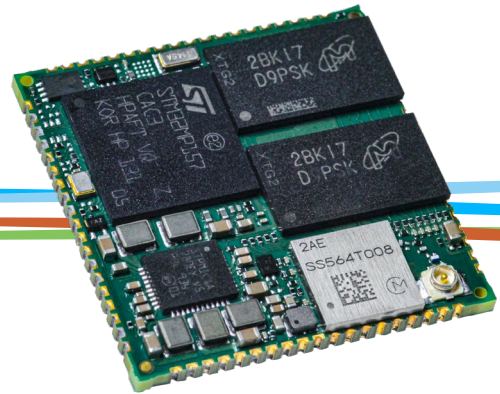




Digi ConnectCore MP1

Intelligent, wireless and secure embedded system-on-module family based on the STM32MP1 MPU with turnkey Linux support in the Digi SMTplus standard form factor

Digi ConnectCore® MP1 modules deliver a highly integrated and secure connected system-on-module solution. The Digi SMTplus® surface-mount form factor provides simplified design integration, efficiency and reliability.

Built on the STM32MP13 and STM32MP15 microprocessors, Digi ConnectCore MP1 modules are the intelligent communication engine for today's secure connected devices with pre-certified dual-band Wi-Fi 5 (802.11a/b/g/n/ac), Bluetooth® 5.2 and Gigabit Ethernet connectivity.

The scalable Digi ConnectCore MP1 module family provides ideal options for headless gateway products with rich connectivity requirements and display devices with advanced HMI's powered by a 3D graphics processing unit (GPU) and camera support. Digi ConnectCore MP1 system-on-modules are designed for industrial and medical use with high levels of security, reliability, performance and 10+ year longevity.

Embedded device security is a critical design aspect for the growing number of connected IoT applications. **Digi ConnectCore SOM solutions** provide built-in security with **Digi TrustFence®**, a fully integrated device-security framework simplifying the process of securing connected devices.

Digi ConnectCore Cloud Services, Digi ConnectCore Security Services and **Expert Support** provide additional capabilities for development, deployment and maintenance.

Digi Embedded Yocto®, Digi's feature-rich Linux distribution with many extensions for embedded product design provides a fully tested, validated and maintained turnkey Linux software platform.

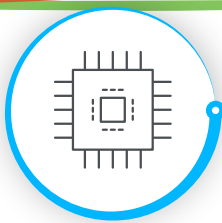
With over twenty years of embedded SOM experience enabling millions of globally connected products, Digi is a trusted embedded and IoT solutions provider, simplifying the way customers design, build and deploy connected applications. **Digi Wireless Design Services** (WDS) offer additional cellular integration support, certification assistance, and custom design and build services to get your products to market smarter and faster.

PMIC	Up to 1 GB SLC NAND	Up to 1 GB DDR3L	Wi-Fi 5 802.11a/b/g/n/ac	Bluetooth 5.2
Digi Microcontroller Assist™ (optional)			Secure Element (optional)	
Arm® Single or Dual* Cortex®-A7 650 MHz L1 32 kB I L1 32 kB D 128 kB / 256 kB L2 Cache			Arm® Cortex®-M4 209 MHz* FPU MPU	
EXTERNAL MEMORY			INTERNAL MEMORY	
DDR3/DDR3L 16/32-bit at 533 MHz			MCU System RAM 384 kB Retention RAM 64 kB	
8-bit SLC NAND 8-bit ECC			System RAM 128/256 kB	
Up to 3x SDMMC			Backup RAM 4/8 kB	OTP fuse 3 kB
CONNECTIVITY			GRAPHICS*	
1x or 2x** 10/100 or Gigabit Ethernet GMAC IEEE 1588v2			3D GPU OpenGL ES 2.0 at 533 MHz	
Up to 3x USB 2.0 Host/OTG with 2x HS PHY			MIPI-DSI controller	
Camera interface*			LCD-TFT controller	
HDMI-CEC*			SYSTEM / CONTROL	
2x CAN FD including 1x TTCAN			LDOs	
Digital Filter Up to 8 channels / 6 filters			Internal and external or crystal oscillators	
Up to 6x SPI Up to 4x I2S			MDMA + DMA	
Up to 6x I2C			Reset and clock	
4x UART 4x USART			Up to 3x watchdogs	
Up to 4x SAI			Up to 176 GPIO	
8x8 keypad			2x 16-bit advanced motor control timers	
ASRC			10x 16-bit all-purpose timers	
SPDIF			2x 32-bit timers, 41C/OC/PWM	
			5x 16-bit low-power timers	
			Temperature sensor	
			SECURITY	
			Digi TrustFence®	
			TrustZone®	
			AES 256, TDES	
			SHA-256, MD5, HMAC*	
			SHA-256/512, SHA-3, HMAC**	
			3x tamper pins 1 active* 12x tamper pins 5 active**	
			Secure boot	
			Secure RAM	
			Secure peripherals	
			Secure RTC	
			Analog true RNG	
			96-bit unique ID	
			ANALOG	
			2x 16-bit ADC* / 2x 12-bit ADC**	
			2x 12-bit DAC*	

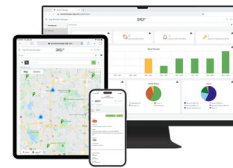
*Digi ConnectCore MP1 only **Digi ConnectCore MP13 only

Key features, benefits and applications

- Industrial-grade, scalable, embedded SOM platform
- Pre-certified dual-band Wi-Fi 5 (802.11ac) and Bluetooth 5.2
- Power management with hardware and software support
- Digi SMTplus form factor (29 x 29 mm) for ultimate reliability
- High level of pin-compatibility with **ConnectCore 6UL SOMs**
- Seamless cellular modem and **Digi XBee®** integration
- **Digi ConnectCore Cloud Services** remote monitoring, device management and IoT application enablement
- **Digi Embedded Yocto** Linux support
- Turnkey development services available from **Digi WDS**



DIGI CONNECTCORE MP1 Specifications



Connect this device with
Digi ConnectCore Cloud Services.
Create. Configure. Deploy. Manage.

SPECIFICATIONS		DIGI CONNECTCORE MP157 SOM	DIGI CONNECTCORE MP133 SOM
FEATURES			
APPLICATION PROCESSOR	STMicroelectronics STM32MP157C, Arm® dual Cortex®-A7 at 650 MHz, Cortex-M4 at 209 MHz, FPU/MPU, 3D GPU, secure boot, crypto engine		STMicroelectronics STM32MP133C, Arm Cortex-A7 at 650 MHz, secure boot, crypto engine
MEMORY	Up to 1 GB SLC NAND flash, up to 1 GB DDR3L		256 MB SLC NAND flash, 256 MB DDR3L
PMIC	STMicroelectronics Power Management IC — STPMIC1		
VIDEO / GRAPHICS	3D GPU (Vivante® — OpenGL® ES 2.0) running at up to 533 MHz, with performance up to 26 Mtriangle/s, 133 Mpixel/s Parallel LCD-TFT controller, up to 24-bit digital RGB888, up to WXGA (1366 × 768) at 60 fps or up to Full HD (1920 × 1080) at 30 fps MIPI® DSI 2 data lanes up to 1 Gbps each, up to WXGA at 60 fps (63 M pixels/s) or up to Full HD (1920 × 1080) at 30 fps		
CAMERA	8- to 14-bit camera		
SECURITY	Secure boot, TrustZone® peripherals, 3x tamper pins including 1x active tamper, 2x TRNG (6 triple oscillators), 2x CRC calculation units; 2x cryptographic processors, hardware acceleration with DMA support; encryption/decryption: DES/TDES: ECB (electronic codebook) and CBC (cipher block chaining) 64-, 128- or 192-bit key; AES: ECB, CBC, GCM, CCM, and CTR (counter mode) chaining algorithms, 128-, 192- or 256-bit key; Universal HASH: SHA-1, SHA-224 and SHA-256 (secure HASH algorithms), MD5, HMAC; Cortex-M4 resources isolation; Digi TrustFence embedded security framework		Secure boot, TrustZone 12x tamper pins incl. 5x active tamper, 1x TRNG (6 triple oscillators), 1x CRC calc. unit, 1x cryptographic processor, hardware acceleration with DMA support; encryption/decryption: DES/TDES: ECB (electronic codebook) and CBC (cipher block chaining) 64-, 128- or 192-bit key; AES: ECB, CBC, GCM, CCM, and CTR (counter mode) chaining algorithms, 128-, 192- or 256-bit key; Universal HASH: SHA-1, SHA-224 and SHA-256, SHA-384, SHA-512, SHA-3, HMAC; Digi TrustFence embedded security framework
PERIPHERALS / INTERFACES	6x I2C, 4x UART + 4x USART, 6x SPI (3 with full duplex I2S); 4x SAI (stereo audio: I2S, PDM, SPDIF Tx), SPDIF Rx (4 inputs); HDMI-CEC; 3x SDMMC up to 8-bit (SD / eMMC™ / SDIO); 2x CAN FD (including 1x TTCAN); 2x USB 2.0 high-speed host + 1x USB 2.0 full-speed OTG; Up to 176 I/O ports with interrupt capability; 2x ADCs (up to 16-bit); 1x temperature sensor; 2x 12-bit D/A converters (1 MHz); 1x Digital Filter for Sigma Delta Modulator (DFSDM) with 8 channels/6 filters, internal or external ADC/DAC reference VREF+; 2x 32-bit timers with up to 4 IC/OC/PWM or pulse counter and quadrature (incremental) encoder input; 2x 16-bit advanced motor control timers; 10x 16-bit general-purpose timers (including 2 basic timers without PWM); 5x 16-bit low-power timers; Secure RTC; 2x 4 Cortex-A7 system timers (secure, nonsecure, virtual, hypervisor); 1x SysTick M4 timer; 3x watchdogs (2x independent and window)		5x I2C, 4x UART + 4x USART, 5x SPI (4 with full duplex I2S); 2x SAI (stereo audio: I2S, PDM, SPDIF Tx), SPDIF Rx (4 inputs); 2x SDMMC up to 8-bit (SD / eMMC / SDIO); 2x CAN FD (including 1x TTCAN); 2x USB 2.0 high-speed host or 1x USB 2.0 high-speed host + 1x USB 2.0 full-speed OTG, up to 135 I/O ports with interrupt capability; 2x ADCs (up to 12-bit), 1x temperature sensor, 1x DFSDM (4 channels/2 filters), internal or external ADC reference VREF+; 2x 32-bit timers with up to 4 IC/OC/PWM or pulse counter and quadrature (incremental) encoder input; 2x 16-bit advanced timers, 10x 16-bit general-purpose timers (including 2 basic timers without PWM); 5x 16-bit low-power timers, Secure RTC; 4 Cortex-A7 system timers (secure, nonsecure, virtual, hypervisor); 2x independent watchdogs
ETHERNET	10/100 M or Gigabit Ethernet GMAC IEEE 1588v2, MII/RMII/GMII/RGMII		2x 10/100 M or Gigabit Ethernet GMAC IEEE 1588v2, MII/RMII/RGMII
WIRELESS	Wi-Fi 5 dual-band 802.11a/b/g/n/ac 1x1 radio (up to 433.3 Mbps) with strong WPA3-Enterprise authentication/encryption -40 °C to 85 °C (-40 °F to 185 °F) full temperature range; Bluetooth 5.2 (Basic Rate, Enhanced Data Rate and Bluetooth Low Energy)		
OPERATING TEMPERATURE	Industrial: -40 °C to 85 °C (-40 °F to 185 °F); depending on use case and enclosure/system design		
STORAGE TEMPERATURE	-50 °C to 125 °C (-58 °F to 257 °F)		
RELATIVE HUMIDITY	5% to 90% (non-condensing)		
RADIO APPROVALS*	US, Canada, EU, Japan, Australia/New Zealand, Brazil, Mexico		
EMISSIONS / IMMUNITY / SAFETY*	FCC Part 15 Class B, EN 55022 Class B, EN 61000-3-2, EN 61000-3-3, ICES-003 Class B, VCCI Class II, AS 3548 FCC Part 15 Subpart C Section 15.247, IC (Industry Canada), RSS-210 Issue 5 Section 6.2.2(o), EN 300 328, EN 301 489-17, EN 55024 EN 301 489-3, Safety (IEC 62368-1); visit digi.com/resources/certifications for latest updates		
DESIGN VERIFICATION*	Temperature: IEC 60068-2-1, IEC 60068-2-2, IEC 60068-2-78; vibration/shock: IEC 60068-2-6, IEC 60068-2-64, IEC 60068-2-27, HALT		
MOUNTING / PIN COUNT	Common Digi SMTplus surface mount footprint using 76-pad edge castellated pads (1.27 mm pitch) or 245-pad LGA (1.27 mm pitch) option		
MECHANICAL DIMENSIONS	29 mm x 29 mm x 3 mm (1.14 in x 1.14 in x 0.12 in)		
PRODUCT WARRANTY	3-year		

*Visit [digi.com/resources/certifications](https://www.digi.com/resources/certifications) for latest updates.



DIGI CONNECTCORE MP1

Part Numbers

PART NUMBERS

DIGI CONNECTCORE MP1 DEVELOPMENT KITS

DIGI CONNECTCORE MP157 DEVELOPMENT KIT

CC-WMP157-KIT	Digi ConnectCore MP157 Development Kit with development board and Digi ConnectCore MP157 dual 512 MB/512 MB wireless SOM
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DIGI CONNECTCORE MP133 DEVELOPMENT KIT

CC-WMP133-KIT	Digi ConnectCore MP133 Development Kit with development board and Digi ConnectCore MP133 256 MB/256 MB wireless SOM
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PART NUMBERS

DIGI CONNECTCORE MP157 SYSTEM-ON-MODULES

DIGI CONNECTCORE MP157 — SECURE WIRELESS MODULE

CC-WST-DW69-NM	Digi ConnectCore MP157 dual 650 MHz, M4, GPU, 512 MB DDR3L, 512 MB SLC NAND flash, secure boot, -40 to 85 °C (-40 to 185 °F), Wi-Fi 5, Bluetooth 5.2
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CC-WST-DW7A-NM	Digi ConnectCore MP157 dual 650 MHz, M4, GPU, 1 GB DDR3L, 1 GB SLC NAND flash, secure boot, -40 to 85 °C (-40 to 185 °F), Wi-Fi 5, Bluetooth 5.2
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DIGI CONNECTCORE MP157 — SECURE ETHERNET MODULE

CC-ST-DW69-ZM	Digi ConnectCore MP157 dual 650 MHz, M4, GPU, 512 MB DDR3L, 512 MB SLC NAND flash, secure boot, -40 to 85 °C (-40 to 185 °F)
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CC-ST-DW7A-ZM	Digi ConnectCore MP157 dual 650 MHz, M4, GPU, 1 GB DDR3L, 1 GB SLC NAND flash, secure boot, -40 to 85 °C (-40 to 185 °F)
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PART NUMBERS

DIGI CONNECTCORE MP133 SYSTEM-ON-MODULES

DIGI CONNECTCORE MP133 — SECURE WIRELESS MODULE

CC-WST-DX58-NK	Digi ConnectCore MP133 single 650 MHz, 256 MB DDR3L, 256 MB SLC NAND flash, secure boot, 2x Gbe, -40 to 85 °C (-40 to 185 °F), Wi-Fi 5, Bluetooth 5.2
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DIGI CONNECTCORE MP133 — SECURE ETHERNET MODULE

CC-ST-DX58-ZK	Digi ConnectCore MP133 single 650 MHz, 256 MB DDR3L, 256 MB SLC NAND flash, secure boot, 2x Gbe, -40 to 85 °C (-40 to 185 °F)
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PART NUMBERS

DIGI CONNECTCORE MP1 ACCESSORIES

ACCESSORIES

CC-ACC-LCDH-10	LCD application kit, including 10 in WXGA (high resolution) LCD panel
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DIGI CONNECTCORE MP1

Part Numbers, Services and Support

PART NUMBERS

DIGI CONNECTCORE CLOUD SERVICES

CC-CS-ESS-1-1YR	Digi ConnectCore Cloud Services — Essential 1-year (1 device)
CC-CS-ESS-100-1YR	Digi ConnectCore Cloud Services — Essential 1-year (100 devices)
CC-CS-ESS-100-3YR	Digi ConnectCore Cloud Services — Essential 3-year (100 devices)
CC-CS-ESS-100-5YR	Digi ConnectCore Cloud Services — Essential 5-year (100 devices)
CC-CS-PRM-1-1YR	Digi ConnectCore Cloud Services — Premium 1-year (1 device)
CC-CS-PRM-100-1YR	Digi ConnectCore Cloud Services — Premium 1-year (100 devices)
CC-CS-PRM-100-3YR	Digi ConnectCore Cloud Services — Premium 3-year (100 devices)
CC-CS-PRM-100-5YR	Digi ConnectCore Cloud Services — Premium 5-year (100 devices)

PART NUMBERS

DIGI CONNECTCORE SECURITY SERVICES

CC-SEC-ESS	Digi ConnectCore Security Services — Essential 1-year
CC-SEC-PRM	Digi ConnectCore Security Services — Premium 1-year
CC-SEC-ENT	Digi ConnectCore Security Services — Enterprise 1-year
CC-SEC-ENT-ADD	Digi ConnectCore Security Services — Additional product (same duration as main contract)

PART NUMBERS

EXPERT SUPPORT

DG-EXPE-12	Expert 12x5 support for end users
DG-EXPE-24	Expert 24x7 support for end users



For more information, please visit digi.com.



For more information about Digi ConnectCore MP1,
visit digi.com/ccmp1.

877-912-3444 | 952-912-3444

